UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 26

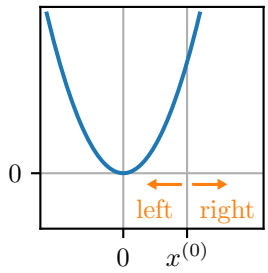
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Example 1. Let $n = 1$ and $f(x) = x^2$. Assume $x^{(0)} = 1$.

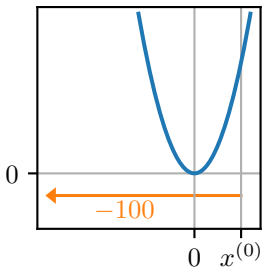


$\Delta x^{(0)}$ determines if we should move to the left or right:

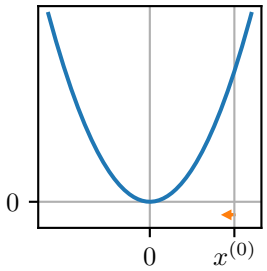
- $\Delta x^{(0)} > 0$ means "move to the right."
- $\Delta x^{(0)} < 0$ means "move to the left."

Note: $\Delta x^{(0)}$ includes numbers like $10, 100, \ldots$ which are all different but correspond to the same direction.
$s^{(0)}$ determines how far we should move along direction $\Delta x^{(0)}$.

- For example, if $\Delta x^{(0)} = -100$ and $s^{(0)} = 1$, we jump over the solution. So $s^{(0)}$ should be small to adjust the value of $x^{(1)}$.



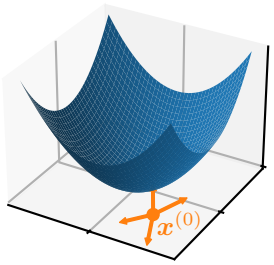
- If $\Delta x^{(0)} = 0.001$ and $s^{(0)} = 1$, $x^{(1)}$ is almost the same as $x^{(0)}$. So $s^{(0)}$ should be big such that $x^{(1)}$ gets much better than $x^{(0)}$.



■

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Example 2. Let $n = 2$ and $f(\boldsymbol{x}) = x_1^2 + x_2^2$.



Given $\boldsymbol{x}^{(0)}$, the direction $\Delta\boldsymbol{x}^{(0)}$ is no longer discrete (left or right) and it can continuously change from north to south. ■

To solve $\min f(\boldsymbol{x})$, assume we designed a descent algorithm:

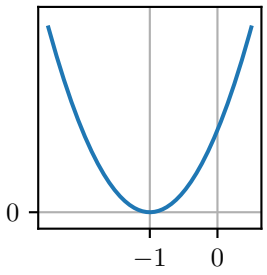
$$\boldsymbol{x}^{(0)} \to \boldsymbol{x}^{(1)} \to \boldsymbol{x}^{(2)} \to \cdots$$

such that

$$f(\boldsymbol{x}^{(0)}) > f(\boldsymbol{x}^{(1)}) > f(\boldsymbol{x}^{(2)}) > \cdots .$$

- The condition
$$f(\boldsymbol{x}^{(k)}) > f(\boldsymbol{x}^{(k+1)}), \quad \forall k \in \{0, 1, 2, \dots\} \tag{1}$$
is not enough to converge to a solution.

- Example: consider $\min_{x\in\mathbb{R}}(x+1)^2$.



Assume $x^{(0)} = 1$, $x^{(1)} = 1/2$, $x^{(2)} = 1/2^2$, …, $x^{(k)} = 1/2^k$. We can see that (1) holds but

$$x^{(0)} \to x^{(1)} \to x^{(2)} \to \cdots \to 0$$

instead of $x^* = -1$.

- We need to make sure that $f(\boldsymbol{x}^{(k+1)}) - f(\boldsymbol{x}^{(k)})$ is negative enough so that there is a good improvement in each iteration k.

Theorem 1. *Assume the objective function $f(\boldsymbol{x})$ is differentiable.*

(i) If $\nabla f(\boldsymbol{x}^{(k)})^{\mathrm{T}}\Delta\boldsymbol{x}^{(k)} < 0$, then $f(\boldsymbol{x}^{(k+1)}) < f(\boldsymbol{x}^{(k)})$ for all small enough step sizes $s^{(k)}$.

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(ii) If $\nabla f(\boldsymbol{x}^{(k)})^{\mathrm{T}}\Delta\boldsymbol{x}^{(k)} > 0$*, then* $f(\boldsymbol{x}^{(k+1)}) > f(\boldsymbol{x}^{(k)})$ *for all small enough step sizes* $s^{(k)}$*.*

(If $\nabla f(\boldsymbol{x}^{(k)})^{\mathrm{T}}\Delta\boldsymbol{x}^{(k)} = 0$*, no conclusion can be made.)*

Due to the above theorem, we should choose a descent direction:

Theorem 2. *Assume the objective function* $f(\boldsymbol{x})$ *is twice differentiable.*

(i) If $\nabla f(\boldsymbol{x}^{(k)}) \neq \mathbf{0}$*, then* $\Delta\boldsymbol{x}^{(k)} = -\nabla f(\boldsymbol{x}^{(k)})$ *is a descent direction.*

(ii) If $\nabla f(\boldsymbol{x}^{(k)}) \neq \mathbf{0}$ *and* $\nabla^2 f(\boldsymbol{x}^{(k)}) \succ \mathbf{0}$*, then* $\Delta\boldsymbol{x}^{(k)} = -\nabla^2 f(\boldsymbol{x}^{(k)})^{-1} \cdot \nabla f(\boldsymbol{x}^{(k)})$ *is a descent direction.*

Proof.

(i) Since $\nabla f(\boldsymbol{x}^{(k)}) \neq \mathbf{0}$,

$$\nabla f(\boldsymbol{x}^{(k)})^{\mathrm{T}}\Delta\boldsymbol{x}^{(k)} = -\|\nabla f(\boldsymbol{x}^{(k)})\|^2 < 0.$$

(ii) Since $\nabla^2 f(\boldsymbol{x}^{(k)}) \succ \mathbf{0}$, we have $\nabla^2 f(\boldsymbol{x}^{(k)})^{-1} \succ \mathbf{0}$ (this is because the eigenvalues of $\mathbf{A}^{-1}$ are the reciprocals of the eigenvalues of $\mathbf{A}$, so $\mathbf{A} \succ \mathbf{0} \Rightarrow \mathbf{A}^{-1} \succ \mathbf{0}$). Therefore,

$$\nabla f(\boldsymbol{x}^{(k)})^{\mathrm{T}}\Delta\boldsymbol{x}^{(k)} = -\underbrace{\nabla f(\boldsymbol{x}^{(k)})^{\mathrm{T}}}_{\boldsymbol{y}^{\mathrm{T}}}\underbrace{\nabla^2 f(\boldsymbol{x}^{(k)})^{-1}}_{\mathbf{A}^{-1}}\underbrace{\nabla f(\boldsymbol{x}^{(k)})}_{\boldsymbol{y}} < 0.$$

■

Note: If $\nabla f(\boldsymbol{x}^{(k)}) = \mathbf{0}$, then $\boldsymbol{x}^{(k)}$ is a stationary point, so we stop the algorithm and don't need $\Delta\boldsymbol{x}^{(k)}$. Hence, the assumption $\nabla f(\boldsymbol{x}^{(k)}) \neq \mathbf{0}$ in the previous theorem is fine because otherwise, we don't need a descent algorithm.
Theorem 2 leads to two algorithms:

- *Gradient algorithm*:
$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} - s^{(k)}\nabla f(\boldsymbol{x}^{(k)}).$$
- *Newton's algorithm*:
$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} - s^{(k)}\nabla^2 f(\boldsymbol{x}^{(k)})^{-1}\nabla f(\boldsymbol{x}^{(k)}).$$

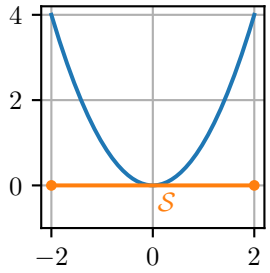
Note: If $\nabla^2 f(\boldsymbol{x}^{(k)})$ is not invertible, Newton's algorithm cannot be used and it should be modified, say add $\lambda\mathbf{I}$ to $\nabla^2 f(\boldsymbol{x}^{(k)})$ for a large λ and use $(\nabla^2 f(\boldsymbol{x}^{(k)}) + \lambda\mathbf{I})^{-1}$. This generates a descent direction but is out of scope for this class.
Let's analyze the gradient algorithm.

- Given an initial guess $\boldsymbol{x}^{(0)}$, define
$$\mathcal{S} = \{\boldsymbol{x} \in \mathbb{R}^n \mid f(\boldsymbol{x}) \leq f(\boldsymbol{x}^{(0)})\}.$$

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- Example: let $f(x) = x^2$ and $x^{(0)} = 2$, then

$$\mathcal{S} = \{x \mid x^2 \le 2^2\} = \{x \mid |x| \le 2\}.$$



- It is said that ∇f is *Lipschitz continuous* on $\mathcal{S}$ with constant $L > 0$ if

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \le L\|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathcal{S}.$$

- Lemma: if f is twice continuously differentiable and $\mathcal{S}$ is compact, then L exists.
- If $\mathcal{S}$ is not compact, the optimization may not have a solution.

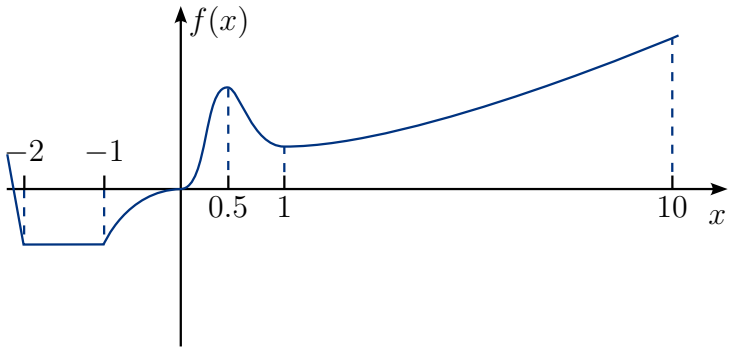
  Example: $\min \mathrm{e}^x$. Assume $x^{(0)} = 0$, then $\mathcal{S} = \{x \mid \mathrm{e}^x \le \mathrm{e}^0 = 1\} = \{x \mid x \le 0\}$ is not compact. Also, x^* does not exist.

Theorem 3. *Consider the gradient algorithm*

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} - s^{(k)} \nabla f(\boldsymbol{x}^{(k)}), \quad k = 0, 1, \ldots.$$

Consider an arbitrary $\varepsilon > 0$. If $s^{(0)}, s^{(1)}, \ldots$ are all chosen in the interval $(\varepsilon, 2/L)$, then $\|\nabla f(\boldsymbol{x}^{(k)})\| \to 0$ as $k \to \infty$.

Note: If $\varepsilon = 0$ and $s^{(k)}$ are chosen to converge to zero, then the algorithm cannot make enough progress. That's why $\varepsilon > 0$.
Illustration:



- At $x = 1$, $x = 0$, $x = 0.5$ and $x \in [-2, -1]$, we have $\|\nabla f(x)\| = |f'(x)| = 0$. By Theorem 3, as $k \to \infty$, $x^{(k)}$ approaches 1, 0, 0.5 or $[-2, -1]$.

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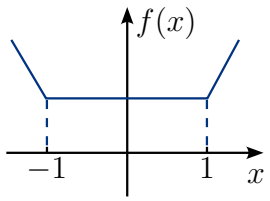
- $x = 0.5$ is a local maximum but we may end up there. For example, if $x^{(0)} = 10$, for a choice of $s^{(0)}$ we have $x^{(1)} = 0.5$ since $f(x^{(1)}) < f(x^{(0)})$ in the above plot. Then $f'(x^{(1)}) = 0$ and $x^{(2)} = x^{(1)} - s^{(1)} f'(x^{(1)}) = x^{(1)}$. Similarly, $x^{(k)} = x^{(1)} = 0.5$, $\forall k \geq 1$.
- $x = 0$ is a saddle point (a stationary point that is neither a minimum nor maximum). We may end up there as well.

If $f(\boldsymbol{x})$ is convex, then $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ iff $\boldsymbol{x}^*$ is a global minimum. Thus, the gradient algorithm for $\min f(\boldsymbol{x})$ with $f(\boldsymbol{x})$ being convex always finds a global minimum if $s^{(k)}$ are small enough. Consider $\min f(\boldsymbol{x})$ where $f(\boldsymbol{x})$ is convex.

(1) If the solution $\boldsymbol{x}^*$ is unique, then

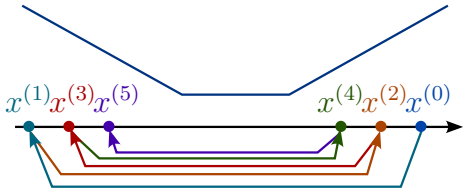
$$\boldsymbol{x}^{(0)} \to \boldsymbol{x}^{(1)} \to \boldsymbol{x}^{(2)} \to \cdots \to \boldsymbol{x}^*. \quad \text{(converges to } \boldsymbol{x}^*\text{)}$$

(2) If the solution is not unique (recall that the set of solutions is convex), for example:



Assume the gradient algorithm generates the following sequence:

$$\underset{1+\frac{1}{2^0}}{\boxed{x^{(0)}}} \to \underset{-1-\frac{1}{2^0}}{\boxed{x^{(1)}}} \to \underset{1+\frac{1}{2^1}}{\boxed{x^{(2)}}} \to \underset{-1-\frac{1}{2^1}}{\boxed{x^{(3)}}} \to \underset{1+\frac{1}{2^2}}{\boxed{x^{(4)}}} \to \underset{-1-\frac{1}{2^2}}{\boxed{x^{(5)}}} \to \cdots .$$



Then we have

$$x^{(k)} \to \begin{cases} 1, & \text{if } k \text{ is even,} \\ -1, & \text{if } k \text{ is odd,} \end{cases}$$

but $1, -1$ are both solutions. Thus in this case, different parts of the sequence $\boldsymbol{x}^{(0)} \to \boldsymbol{x}^{(1)} \to \cdots$ may converge to different value.

Step size of the gradient algorithm:

- The easiest type of step size is the constant step size:

$$s^{(0)} = s^{(1)} = \cdots = s.$$

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- If $0 < s < 2/L$, the gradient algorithm works (gradient converges to zero).

Topics out of scope for this class:

- Newton's algorithm also finds a point $\boldsymbol{x}^*$ such that $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ if $s^{(0)}, s^{(1)}, \ldots$ are small.
- Convergence rate: Newton's method converges faster than the gradient algorithm.
- Algorithm design when there are constraints.

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